

PRACTICE SHEET-1

1. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is:
 (1) $8\sqrt{2}$ cm (2) 16 cm
 (3) $\sqrt{32}$ cm (4) 8 cm
2. The breadth of a rectangular hall is three-fourth of its length. If the area of the floor is 768 sq. m., then the difference between the length and breadth of the hall is:
 (1) 8 metres (2) 12 metres
 (3) 24 metres (4) 32 metres
3. The length of a plot is five times its breadth. A playground measuring 245 square metres occupies half of the total area of the plot. What is the length of the plot?
 (1) $35\sqrt{2}$ metres (2) $175\sqrt{2}$ metres
 (3) 490 metres (4) $5\sqrt{2}$ metres
4. The difference between the length and breadth of a rectangle is 23 m. If its perimeter is 206 m, then its area is
 (1) 1520 m^2 (2) 2420 m^2
 (3) 2480 m^2 (4) 2520 m^2
5. A path of uniform width runs round the inside of a rectangular field 38 m long and 32 m wide. If the path occupies 600 m^2 , then the width of the path is
 (1) 30 m (2) 5 m
 (3) 18.75 m (4) 10 m
6. The length and breadth of a rectangle are increased by 20% and 25% respectively. The increase in the area of the resulting rectangle will be:
 (1) 60% (2) 50%
 (3) 40% (4) 30%
7. The length of a room floor exceeds its breadth by 20 m. The area of the floor remains unaltered when the length is decreased by 10 m but the breadth is increased by 5 m. The area of the floor (in square metres) is :
 (1) 280 (2) 325
 (3) 300 (4) 420
8. The sides of a triangle are 3 cm, 4 cm and 5 cm. The area (in cm^2) of the triangle formed by joining the mid points of this triangle is:
 (1) 6 (2) 3
 (3) $3/2$ (4) $3/4$
9. From a point in the interior of an equilateral triangle, the length of the perpendiculars to the three sides are 6 cm, 8 cm and 10 cm respectively. The area of the triangle is
 (1) 48 cm^2 (2) $16\sqrt{3}$ cm^2
 (3) $192\sqrt{3}$ cm^2 (4) 192 cm^2
10. The area of an isosceles triangle is 4 square unit. If the length of the third side is 2 unit, the length of each equal side is
 (1) 4 units (2) $2\sqrt{3}$ units
 (3) $\sqrt{17}$ units (4) $3\sqrt{2}$ units
11. 360 sq. cm and 250 sq. cm are the area of two similar triangles. If the length of one of the sides of the first triangle be 8 cm, then the length of the corresponding side of the second triangle is
 (1) $6\frac{1}{5}$ cm (2) $6\frac{1}{3}$ cm
 (3) $6\frac{2}{3}$ (4) 6 cm
12. The perimeter of an isosceles triangle is 544 cm and each of the equal sides is $\frac{5}{6}$ times the base. What is the area (in cm^2) of the triangle?
 (1) 38172 (2) 18372
 (3) 31872 (4) 13872
13. The perimeter of a rhombus is 100 cm. If one of its diagonals is 14 cm, then the area of the rhombus is
 (1) 144 cm^2 (2) 225 cm^2
 (3) 336 cm^2 (4) 400 cm^2
14. If the measure of one side and one diagonal of a rhombus are 10 cm and 16 cm respectively, then its area (in cm^2) is :
 (1) 60 (2) 64
 (3) 96 (4) 100

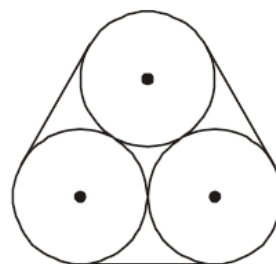
15. The ratio of the length of the parallel sides of a trapezium is 3:2. The shortest distance between them is 15 cm. If the area of the trapezium is 450 cm^2 , the sum of the length of the parallel sides is
 (1) 15 cm (2) 36 cm
 (3) 42 cm (4) 60 cm
16. A parallelogram has sides 15 cm and 7 cm long. The length of one of the diagonals is 20 cm. The area of the parallelogram is
 (1) 42 cm^2 (2) 60 cm^2
 (3) 84 cm^2 (4) 96 cm^2
17. In $\triangle ABC$, D and E are the points of sides AB and BC respectively such that $DE \parallel AC$ and $AD : DB = 3 : 2$. The ratio of area of trapezium ACED to that of $\triangle BED$ is
 (1) 4 : 15 (2) 15 : 4
 (3) 4 : 21 (4) 21 : 4
18. ABCD is a trapezium in which $AB \parallel DC$ and $AB = 2 CD$. The diagonals AC and BD meet at O. The ratio of area of triangles AOB and COD is
 (1) 1 : 1 (2) $1 : \sqrt{2}$
 (3) 4 : 1 (4) 1 : 4
19. The areas of a square and a rectangle are equal. The length of the rectangle is greater than the length of any side of the square by 5 cm and the breadth is less by 3 cm. Find the perimeter of the rectangle.
 (1) 17 cm (2) 26 cm
 (3) 30 cm (4) 34 cm
20. If the side of a square is increased by 25%, then its area is increased by:
 (1) 25% (2) 55%
 (3) 40.5% (4) 56.25%
21. ABC is an isosceles right angled triangle with $\angle B = 90^\circ$. On the sides AC and AB, two equilateral triangles ACD and ABE have been constructed. The ratio of area of $\triangle ABE$ and $\triangle ACD$ is
 (1) 1 : 3 (2) 2 : 3
 (3) 1 : 2 (4) 1 : 2
22. Two triangles ABC and DEF are similar to each other in which $AB = 10 \text{ cm}$, $DE = 8 \text{ cm}$. Then the ratio of the area of triangles ABC and DEF is
 (1) 4 : 5 (2) 25 : 16
 (3) 64 : 125 (4) 4 : 7
23. If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3 \text{ cm}$, $EF = 4 \text{ cm}$ and area of $\triangle ABC = 54 \text{ cm}^2$, then the area of $\triangle DEF$ is :
 (1) 66 cm^2 (2) 78 cm^2
 (3) 96 cm^2 (4) 54 cm^2
24. In $\triangle ABC$, O is the centroid and AD, BE, CF are three medians and the area of $\triangle AOE = 15 \text{ cm}^2$, then area of quadrilateral BDOF is
 (1) 20 cm^2 (2) 30 cm^2
 (3) 40 cm^2 (4) 25 cm^2
25. A straight line parallel to the base BC of the triangle ABC intersects AB and AC at the points D and E respectively. If the area of the $\triangle ABE$ be 36 sq.cm , then the area of the $\triangle ACD$ is
 (1) 18 sq.cm (2) 36 sq.cm
 (3) 18 cm (4) 36 cm
26. ABC is a right angled triangle, B being the right angle. Mid-points of BC and AC are respectively B' and A'. The ratio of the area of the quadrilateral AA'B'B' to the area of the triangle ABC is
 (1) 1 : 2 (2) 2 : 3
 (3) 3 : 4 (4) N. A.
27. Two triangles ABC and PQR are congruent. If the area of $\triangle ABC$ is 60 sq. cm , then area of $\triangle PQR$ will be
 (1) 60 sq.cm (2) 30 sq.cm
 (3) 15 sq.cm (4) 120 sq.cm
28. Two sides of a parallelogram are 20 cm and 25 cm. If the altitude corresponding to the side of length 25 cm is 10 cm, then the altitude corresponding to the other pair of sides is
 (1) 10.5 cm (2) 12 cm
 (3) 12.5 cm (4) 10 cm
29. The diagonal of a quadrilateral shaped field is 24m and the perpendiculars dropped on it from the remaining opposite vertices are 8m and 13m. The area of the field is
 (1) 252 m^2 (2) 156 m^2
 (3) 96 m^2 (4) 1152 m^2
30. Two isosceles triangles have equal vertical angles and their areas are in the ratio 9:16. Then the ratio of their corresponding heights is-
 (1) 4.5 : 8 (2) 4 : 3
 (3) 8 : 4.5 (4) 3 : 4

ANSWER KEY

1.	(4)	2.	(1)	3.	(1)	4.	(4)	5.	(2)	6.	(2)	7.	(3)	8.	(3)	9.	(3)	10.	(3)
11.	(3)	12.	(4)	13.	(3)	14.	(3)	15.	(4)	16.	(3)	17.	(4)	18.	(3)	19.	(4)	20.	(4)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(2)	26.	(3)	27.	(1)	28.	(3)	29.	(1)	30.	(4)

PRACTICE SHEET-2

- The area of the ring between two concentric circles, whose circumference are 88 cm and 132 cm, is:
 (1) 780 cm² (2) 770 cm²
 (3) 715 cm² (4) 660 cm²
- Four equal circles each of radius 'a' units touch one another. The area enclosed between them ($\pi = 22/7$), in square units, is
 (1) $3a^2$ (2) $\frac{6}{7}a^2$
 (3) $\frac{41}{7}a^2$ (4) $\frac{1}{7}a^2$
- The area of circle whose radius is 6 cm is trisected by two concentric circles. The radius of the smallest circle is
 (1) $2\sqrt{3}$ cm (2) $2\sqrt{6}$ cm
 (3) 2 cm (4) 3 cm
- The radius of circle A is twice that of circle B and the radius of circle B is twice that of circle C. Their area will be in the ratio
 (1) 16 : 4 : 1 (2) 4 : 2 : 1
 (3) 1 : 2 : 4 (4) 1 : 4 : 16
- The four equal circles of radius 4 cm drawn on the four corners of a square touch each other externally. Then the area of the portion between the square and the four sectors is
 (1) $9(\pi - 4)$ sq. cm.
 (2) $16(\pi - 4)$ sq. cm.
 (3) $9(4 - \pi)$ sq. cm.
 (4) $16(4 - \pi)$ sq. cm.
- The area of a circle is proportional to the square of its radius. A small circle of radius 3 cm is drawn within a larger circle of radius 5 cm. Find the ratio of the area of the annular zone to the area of the larger circle. (Area of the annular zone is the difference between the area of the larger circle and that of the smaller circle).
 (1) 9 : 16 (2) 9 : 25
 (3) 16 : 25 (4) 16 : 27
- The diameter of two circles are the side of a square and the diagonal of the square. The ratio of the area of the smaller circle and the larger circle is
 (1) 1 : 2 (2) 1 : 4
 (3) $\sqrt{2} : \sqrt{3}$ (4) 1 : $\sqrt{2}$
- The area of the largest circle, that can be drawn inside a rectangle with sides 18 cm. by 14 cm, is
 (1) 49 cm² (2) 154 cm²
 (3) 378 cm² (4) 1078 cm²
- A circle is inscribed in an equilateral triangle of side 8 cm. The area of the portion between the triangle and the circle is
 (1) 11 cm² (2) 10.95 cm²
 (3) 10 cm² (4) 10.50 cm²
- If the difference between areas of the circum-circle and the in-circle of an equilateral triangle is 44 cm², then the area of the triangle is
 (1) 28 cm² (2) $7\sqrt{3}$ cm²
 (3) $14\sqrt{3}$ cm² (4) 21 cm²
- Three circles of diameter 10 cm each, are bound together by a rubber band, as shown in the figure.



The length of the rubber band, (in cm) if it is stretched as shown, is

- (1) 30 (2) $30 + 10\pi$
(3) 10π (4) $60 + 20\pi$

12. A circular road runs around a circular ground. If the difference between the circumference of the outer circle and the inner circle is 66 metres, the width of the road is:

- (1) 10.5 metres (2) 7 metres
(3) 5.25 metres (4) 21 metres

13. A person observed that he required 30 seconds less time to cross a circular ground along its diameter than to cover it once along the boundary. If his speed was 30 m/minute, then the radius of the circular ground.

- (1) 5.5 m (2) 7.5 m
(3) 10.5 m (4) 3.5 m

14. If the four equal circles of radius 3 cm touch each other externally, then the area of the region bounded by the four circles is

- (1) $4(9 - \pi)$ sq.cm (2) $9(4 - \pi)$ sq.cm
(3) $5(6 - \pi)$ sq.cm (4) $6(5 - \pi)$ sq.cm

15. The area of a circle is increased by 22 cm^2 when its radius is increased by 1 cm. The original radius of the circle is

- (1) 3 cm (2) 5 cm
(3) 7 cm (4) 9 cm

16. The radii of two circles are 5cm and 12cm. The area of a third circle is equal to the sum of the area of the two circles. The radius of the third circle is :

- (1) 13 cm (2) 21 cm
(3) 30 cm (4) 17 cm

17. The perimeter of a semicircular path is 36 m. Find the area of this semicircular path.

- (1) 42sq.m (2) 54 sq. m
(3) 63 sq.m (4) 77 sq. m

18. The ratio between the area of two circles is 4 : 7. What will be the ratio of their radii?

- (1) $2 : \sqrt{7}$ (2) $4 : 7$
(3) $16 : 49$ (4) $4 : \sqrt{7}$

19. Three circles of radius a, b, c touch each other externally. The area of the triangle formed by joining their centre is

- (1) $\sqrt{(a+b+c)abc}$

(2) $(a + b + c) \sqrt{ab + bc + ca}$

(3) $ab + bc + ca$

(4) None of the above

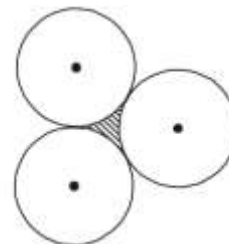
20. The area of a circle is proportional to the square of its radius. A small circle of radius 3 cm is drawn within a larger circle of radius 5 cm. Find the ratio of the area of the annular zone to the area of the larger circle. (Area of the annular zone is the difference between the area of the larger circle and that of the smaller circle).

- (1) 9 : 16 (2) 9 : 25
(3) 16 : 25 (4) 16 : 27

21. The diameter of two circles are the side of a square and the diagonal of the square. The ratio of the area of the smaller circle and the larger circle is

- (1) 1 : 2 (2) 1 : 4
(3) 2 : 3 (4) 1 : 2

22. Three circles of equal radius ' a ' cm touch each other. The area of the shaded region is:



(a) $\left(\frac{\sqrt{3} + \pi}{2} \right) a^2$

(b) $\left(\frac{6\sqrt{3} - \pi}{2} \right) a^2$

(c) $(\sqrt{3} - \pi) a^2$

(d) $\left(\frac{2\sqrt{3} - \pi}{2} \right) a^2$

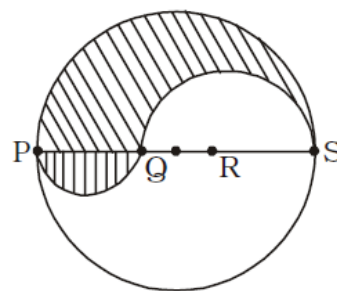
23. The radii of two circles are 10 cm and 24 cm. The radius of a circle whose area is the sum of the area of these two circles is

- (1) 36 cm (2) 17 cm
(3) 34 cm (4) 26 cm

24. The perimeter of a rhombus is 60 cm and one of its diagonal is 24 cm. The area of the rhombus is

- (1) 108 sq. cm. (2) 216 sq. cm.
(3) 432 sq. cm. (4) 206 sq. cm.

25. The ratio of circumference and diameter of a circle is $22 : 7$. If the circumference be $1\frac{4}{7}$ m, then the radius of the circle is :
- (1) $\frac{1}{3}$ m (2) $\frac{1}{2}$ m
(3) $\frac{1}{4}$ m (4) 1 m
26. Four circles of equal radii are described about the four corners of a square so that each touches two of the other circles. If each side of the square is 140 cm then area of the space enclosed between the circumference of the circle is Take
- (1) 4200 cm² (2) 2100 cm²
(3) 7000 cm² (4) 2800 cm²
27. The perimeter of a triangle is 67 cm. The first side is twice the length of the second side. The third side is 11 cm more than the second side. Find the length of the shortest side of the triangle.
- (1) 12 cm. (2) 14 cm.
(3) 17 cm. (4) 25 cm.
28. The radius of a wheel is 25 cm. How many rounds it will take to complete 11 km.
- (1) 5000 (2) 6000
(3) 7000 (4) 4000
29. If the perimeter of circle A is equal to perimeter of semi circle B, what is the ratio of their areas?
- (1) $(\pi + 2)^2 : 2\pi^2$ (2) $2\pi^2 : (\pi + 2)^2$
(3) $(\pi + 2)^2 : 4\pi^2$ (4) $4\pi^2 : (\pi + 2)^2$
30. PS is a diameter of a circle of radius 6 cm. In the diameter PS, Q and R are two points such that PQ, QR and RS are all equal. Semicircles are drawn on PQ and QS as diameter (as shown in the figure). The perimeter of shaded portion is :



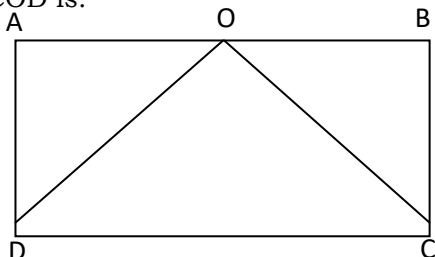
- (1) $15\frac{6}{7}$ cm (2) $75\frac{3}{7}$
(3) $37\frac{5}{7}$ cm (4) $18\frac{6}{7}$ cm

ANSWER KEY

1.	(2)	2.	(2)	3.	(1)	4.	(1)	5.	(4)	6.	(3)	7.	(1)	8.	(2)	9.	(2)	10.	(3)
11.	(2)	12.	(1)	13.	(4)	14.	(2)	15.	(1)	16.	(1)	17.	(4)	18.	(1)	19.	(1)	20.	(3)
21.	(1)	22.	(4)	23.	(4)	24.	(2)	25.	(3)	26.	(1)	27.	(2)	28.	(3)	29.	(3)	30.	(4)

CDS PYQ

1. In the figure given below, the area of rectangle ABCD is 100 sq. cm. O is any point on AB and CD = 20cm. Then, the area of $\triangle COD$ is:



[CDS 2013(I)]

- (a) 40 sq cm (b) 45 sq cm
(c) 50 sq cm (d) 80 sq cm

2. If an isosceles right angled triangle has area 1 sq unit, then what is its perimeter?

[CDS 2013(I)]

- (a) 3 units (b) $2\sqrt{2}+1$ units
(c) $(\sqrt{2}+1)$ units (d) $2(\sqrt{2}+1)$ units

3. A circular water fountain 6.6 m in diameter is surrounded outside by a path of width 1.5m. The area of this path (in sq m) is:

[CDS 2013(I)]

- (a) 13.62π (b) 13.15π
(c) 12.15π (d) None of these

4. The area of a rectangular field is 4500 sq m. If its length and breadth are in the ratio 9:5, then its perimeter is:

[CDS 2013(I)]

- (a) 90m (b) 150m
(c) 280m (d) 360m

5. The area of a square inscribed in a circle of radius 8cm is:

[CDS 2013(I)]

- (a) 32 sq cm (b) 64 sq cm
(c) 128 sq cm (d) 256 sq cm

6. The short and long hands of a clock are 4 cm and 6 cm long, respectively. Then, the ratio of distances travelled by tips of short hand in 2 days and long hand in 3 days is:

[CDS 2013(I)]

- (a) 4:9 (b) 2:9
(c) 2:3 (d) 1:27

7. The arc AB of the circle with centre at O and radius 10cm has length 16cm. What is the area of the sector bounded by the radii OA, OB and the arc AB?

[CDS 2013(I)]

- (a) 40π sq cm (b) 40 sq cm
(c) 80 sq cm (d) 20π sq cm

8. The minute hand of a watch is 2.5 cm long. The distance its extreme end transverses in 40 min is:

[CDS 2013(I)]

- (a) $10\pi/3$ cm (b) $3\pi/10$ cm
(c) $10/3$ cm (d) 10cm

9. If the area of a regular hexagon is $96\sqrt{3}$ sq cm, then its perimeter is:

[CDS 2013(I)]

- (a) 36cm (b) 48cm
(c) 54cm (d) 64cm

10. What is the area of a circle whose area is equal to that of a triangle with sides 7cm, 24cm and 25cm?

[CDS 2013(II)]

- (a) 80cm^2 (b) 84cm^2
(c) 88cm^2 (d) 90cm^2

11. If the area of an equilateral triangle is x and its perimeter is y, then which one of the following is correct?

[CDS 2013(II)]

- (a) $y^4 = 432x^2$ (b) $y^4 = 21x^2$
(c) $y^2 = 432x^2$ (d) None of these

12. The area of an isosceles $\triangle ABC$ with $AB = AC$ and altitude $AD = 3$ cm is 12 sq cm. What is its perimeter?

[CDS 2013(II)]

- (a) 18cm (b) 16cm
(c) 14cm (d) 12cm

13. How many 200mm lengths can be cut from 10m of ribbon?

[CDS 2013(II)]

- (a) 50 (b) 40
(c) 30 (d) 20

14. What is the area between a square of side 10 cm and two inverted semi-circular, cross sections each of radius 5cm inscribed in the square?

[CDS 2013(II)]

- (a) 17.5cm^2 (b) 18.5cm^2
(c) 20.5cm^2 (d) 21.5cm^2

15. The perimeter of a rectangle having area equal to 144cm^2 and sides in the ratio 4 : 9 is:

[CDS 2013(II)]

- (a) 52cm (b) 56cm
(c) 60cm (d) 64cm

16. One side of a parallelogram is 8.06 cm and its perpendicular distance from opposite side is 2.08 cm. What is the approximate area of the parallelogram?

[CDS 2013(II)]

- (a) 12.56cm^2 (b) 14.56cm^2
(c) 16.76cm^2 (d) 22.56cm^2

17. If the diagonals of a rhombus are 4.8 cm and 1.4 cm, then what is the perimeter of the rhombus?

[CDS 2013(II)]

- (a) 5cm (b) 10cm
(c) 12cm (d) 20cm

18. A regular hexagon is inscribed in a circle of radius 5cm. If x is the area inside the circle but outside the regular hexagon, then which one of the following is correct?

[CDS 2013(II)]

- (a) $13\text{cm}^2 < x < 15\text{cm}^2$
(b) $15\text{cm}^2 < x < 17\text{cm}^2$
(c) $17\text{cm}^2 < x < 19\text{cm}^2$
(d) $19\text{cm}^2 < x < 21\text{cm}^2$

19. The area of a rectangle lies between 40cm^2 and 45cm^2 . If one of the sides is 5 cm, then its diagonal lies between:

[CDS 2014(I)]

- (a) 8 cm and 10cm
(b) 9 cm and 11cm
(c) 10 cm and 12 cm
(d) 11 cm and 13 cm

20. How many circular plates of diameter d be taken out of a square plate of side $2d$ with minimum loss of material?

[CDS 2014(I)]

- (a) 8 (b) 6
(c) 4 (d) 2

21. What is the total area of three equilateral triangles inscribed in a semi-circle of radius 2cm?

[CDS 2014(I)]

- (a) 12cm^2 (b) $\frac{3\sqrt{3}}{4}\text{cm}^2$
(c) $\frac{9\sqrt{3}}{4}\text{cm}^2$ (d) $3\sqrt{3}\text{cm}$

22. The area of sector of a circle of radius 36cm is $72\pi\text{cm}^2$. The length of the corresponding arc of the sector is:

[CDS 2014(I)]

- (a) πcm (b) $2\pi\text{cm}$
(c) $3\pi\text{cm}$ (d) $4\pi\text{cm}$

23. A square is inscribed in a circle of diameter $2a$ and another square is circumscribing circle. The difference between the area of outer and inner squares is:

[CDS 2014(I)]

- (a) a^2 (b) $2a^2$
(c) $3a^2$ (d) $4a^2$

24. ABC is a triangle right angled at A. $AB = 6\text{cm}$ and $AC = 8\text{cm}$. Semi-circles drawn (outside the triangle) on AB, AC and BC as diameters which enclose areas x , y square units, respectively. What is $x + y - z$ equal to?

[CDS 2014(I)]

- (a) 48cm^2 (b) 32cm^2
(c) 0 (d) None of these

25. Consider an equilateral triangle of a side of unit length. A new equilateral triangle is formed by joining the mid points of one,

then a third equilateral triangle is formed by joining the mid points of seconds. The process is continued. The perimeter of all triangles, thus formed is:

[CDS 2014(I)]

- (a) 2 units (b) 3 units
(c) 6 units (d) Infinity

26. What is the area of the larger segment of a circle formed by a chord of length 5cm subtending an angle of 90° at the centre?

[CDS 2014(I)]

- (a) $\frac{25}{4} \left(\frac{\pi}{2} + 1 \right) \text{cm}^2$ (b) $\frac{25}{4} \left(\frac{\pi}{2} - 1 \right)$
(c) $\frac{25}{4} \left(\frac{3\pi}{2} + 1 \right)$ (d) None of these

27. A rectangle of maximum area of drawn inside a circle of diameter 5cm. What is the maximum area of such a rectangle?

[CDS 2014(I)]

- (a) 25cm^2 (b) 12.5cm^2
(c) 12cm^2 (d) None of these

28. If AB and CD are two diameters of a circle of radius r and they are mutually perpendicular, then what is the ratio of the area of the circle to the area of the $\triangle ACD$?

[CDS 2014(I)]

- (a) $\frac{\pi}{2}$ (b) π
(c) $\frac{\pi}{4}$ (d) 2π

29. If every side of an equilateral triangle is doubled, then the area of new triangle becomes k times the area of the old one. What is k equal to?

[CDS 2014(II)]

- (a) $\sqrt{3}$ (b) 2
(c) 4 (d) 8

30. A railroad curve is to be laid on a circle. What radius (approx) should be used. If the track is to change direction by 25° in a distance of 120m?

[CDS 2014(II)]

- (a) 300 m (b) 280 m
(c) 275 m (d) 264 m

31. The sides of a triangular field are 41m, 40m and 9m. The number of rose bend that can be prepared in the field if each rose bed, on an average, needs 900 square cam space, is:

[CDS 2015(I)]

- (a) 2000 (b) 1800
(c) 900 (d) 800

32. The ratio of the outer and inner perimeters of a circular path is 23:22. If the path is 5m wide, the diameter of the inner circle is:

[CDS 2015(I)]

- (a) 55m (b) 110m
(c) 220m (d) 230m

33. For equal-sized maximum circular plate are cut off from a square paper sheet of area 784 square cm. The circumference of each plate is:

[CDS 2015(I)]

- (a) 11cm (b) 22cm
(c) 33cm (d) 44cm

34. A boy is cycling such that the wheels of the cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is:

[CDS 2015(II)]

- (a) 15.5km/hour (b) 15.84km/hour
(c) 16km/hour (d) 16.36km/hour

35. There are 437 fruit plants in an orchard planted in rows. The distance between any two adjacent rows in 2m and the distance between any two adjacent plants in 2m. Each rows has the same number of plants. There is 1m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs100 per meter.

[CDS 2015(II)]

- (a) Rs 15,600
(b) Rs 16,800
(c) Rs 18,200
(d) More information is required

36. The circumference of a circle is 100cm. The side of the square inscribed in the circle is:

[CDS 2015(II)]

- (a) $50\sqrt{2}$ cm (b) $\frac{100}{\pi}$ cm
(c) $\frac{50\sqrt{2}}{\pi}$ cm (d) $\frac{100\sqrt{2}}{\pi}$ cm

37. The diameter of a wheel that makes 452 revolutions to move 2km and 26 decameter is equal to:

[CDS 2015(II)]

- (a) $1\frac{9}{22}m$ (b) $1\frac{13}{22}m$
(c) $2\frac{5}{11}m$ (d) $2\frac{7}{11}m$

38. A square is inscribed in a right triangle with legs x and y and has common right angle with the triangle. The perimeter of the square is given by:

[CDS 2015(II)]

- (a) $\frac{2xy}{x+y}$ (b) $\frac{4xy}{x+y}$
(c) $\frac{2xy}{\sqrt{x^2+y^2}}$ (d) $\frac{4xy}{\sqrt{x^2+y^2}}$

39. The area of a trapezium is 336 cm^2 . If its parallel sides are in the ratio 5:7 and the perpendicular distance between them is 14cm, then the smaller of the parallel sides is:

[CDS 2015(II)]

- (a) 20cm (b) 22cm
(c) 24cm (d) 26cm

40. ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by:

[CDS 2015(II)]

- (a) 50% (b) 52%
(c) 54% (d) 56%

41. From a rectangular sheet of sides 18 cm and 14cm, a semicircular portion with smaller side as diameter is taken out. Then the area of the remaining sheet will be:

[CDS 2015(II)]

- (a) 98cm^2 (b) 100cm^2
(c) 108cm^2 (d) 175cm^2

42. A square and an equilateral triangle have equal perimeter. If the diagonal of the square is $12\sqrt{2} \text{ cm}$, then the area of the triangle is:

[CDS 2015(II)]

- (a) $24\sqrt{2}$ (b) $24\sqrt{3}$
(c) $48\sqrt{3}$ (d) $64\sqrt{3}$

43. The number of rounds that a wheel of diameter $7/11$ meter will make in traversing 4 km will be:

[CDS 2016(I)]

- (a) 500 (b) 1000
(c) 1700 (d) 2000

44. The base of an isosceles triangle is 300 unit and each of its equal sides is 170 units. Then the area of the triangle is:

[CDS 2016(I)]

- (a) 9600 square units
(b) 1000 square units
(c) 12000 square units
(d) None of the above

45. Four equal discs are placed such that each one touches two others. If the area of empty space enclosed by them is $150/847$ square centimeter, then the radius of each disc is equal to:

[CDS 2016(I)]

- (a) $7/6 \text{ cm}$ (b) $5/6 \text{ cm}$
(c) $1/2 \text{ cm}$ (d) $5/11 \text{ cm}$

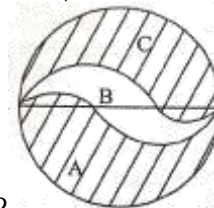
46. A square is inscribed in a right-angled triangle with legs p and q, and has a common right angle with the triangle. The diagonal of the square is given by:

[CDS 2016(I)]

- (a) $\frac{pq}{p+2q}$ (b) $\frac{pq}{2p+q}$
(c) $\frac{\sqrt{2}pq}{p+q}$ (d) $\frac{2pq}{p+q}$

47. A circle of 3m radius is divided into three areas by semicircles of radii 1m and 2m as shown in the figure above. The ratio of the three areas A, B and C will be:

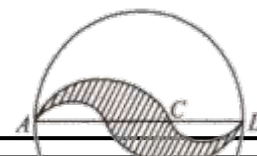
[CDS 2016(I)]



- (a) 2:3:2 (b) 1:1:1
(c) 4:3:4 (d) 1:2:1

48. AD is the diameter of a circle with area 707 m^2 and $AB = BC = CD$ as shown in the figure above. All curves inside the circle are semicircles with their diameters on AD. What is the cost of leveling the shaded region at the rate of 63 per square meter?

[CDS 2016(I)]



- (a) 29,700
(b) 22,400
(c) 14,847
(d) None of the above

49. If a square of side x and an equilateral triangle of side y are inscribed in a circle, then what is the ratio of x to y ?

[CDS 2016(II)]

- (a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{3}{2}}$
(c) $\frac{3}{\sqrt{2}}$ (d) $\frac{\sqrt{2}}{3}$

50. A circle and a square have the same perimeter. Which one of the following is correct?

[CDS 2016(II)]

- (a) The areas are equal
(b) The area of the circle is larger
(c) The area of the square is $\frac{\pi}{2}$ times area of circle
(d) The area of the square is π times area of circle

51. What is the area of a triangle with sides of length 12cm, 13cm and 5cm?

[CDS 2016(II)]

- (a) 30cm^2 (b) 35cm^2
(c) 40cm^2 (d) 42cm^2

52. What is area of largest triangle inscribed in a semi circle of radius r units?

[CDS 2016(II)]

- (a) r^2 square units (b) $2r^2$ square units
(c) $3r^2$ square unit (d) $4r^2$ square unit

53. The diameter of the front wheel of an engine is $2x$ cm and that of rear wheel is $2y$ cm. To cover the same distance, what is the number of times the rear wheel revolves when front wheel revolves n times?

[CDS 2016(II)]

- (a) n/xy (b) ny/x
(c) nx/y (d) xy/n

54. What is the number of round that a wheel of diameter $\frac{5}{11}$ m will make in traversing 7km?

[CDS 2016(II)]

- (a) 3300 (b) 3500
(c) 4400 (d) 4900

55. If a quadrilateral has an inscribed circle, then the sum of a pair of opposite sides equal.

[CDS 2016(II)]

- (a) Half the sum of the diagonals
(b) Sum of the other pair of opposite sides
(c) Sum of two adjacent sides
(d) None of the above

56. The wheel of a car are of diameter 80cm each. The car is travelling at a speed of 66km/hr. What is the number of complete revolutions each wheel makes in 10 minutes?

[CDS 2016(II)]

- (a) 4275 (b) 4350
(c) 4375 (d) 4450

57. If the perimeter of a circle is equal to that of a square, then what is the ratio of area of circle to that of square?

[CDS 2016(II)]

- (a) 22:7 (b) 14:11
(c) 7:22 (d) 11:14

58. If the perimeter of a rectangle is 10 cm and the area is 4cm^2 , then its length is:

[CDS 2017(I)]

- (a) 6cm (b) 5cm
(c) 4.5cm (d) 4cm

59. The areas of two circular field are in the ratio 16:49. If the radius of the bigger field is 14m, then what is the radius of the smaller field?

[CDS 2017(I)]

- (a) 4m (b) 8m
(c) 9m (d) 10m

60. The area of a regular hexagon of side 'a' is equal to:

[CDS 2017(I)]

- (a) $\frac{\sqrt{2}}{3} a^2$ square units
(b) $\frac{3\sqrt{3}}{2} a^2$ square units
(c) $\frac{1}{3} a^2$ square units
(d) $\frac{\sqrt{3}}{2} a^2$ square units

61. If each of the dimensions of a rectangle is increased by 200%, the area is increased by:

[CDS 2017(I)]

- (a) 300% (b) 400%

- (c) 600% (d) 800%

- 62.** Three circles each of radius 3.5 cm touch one another. The area subtended between them is:

[CDS 2017(I)]

- (a) $6(\sqrt{3}\pi - 2)$ (b) $6(2\pi - \sqrt{3}\pi)$
(c) $\frac{49}{8}(2\sqrt{3} - \pi)$ (d) $\frac{49}{8}(\sqrt{3} - \pi)$

- 63.** A copper wire when bent in the form of a square encloses an area of 121 cm. If the same wire is bent in the form of a circle, it encloses an area equal to

[CDS 2017(I)]

- (a) 121 cm² (b) 144 cm²
(c) 154 cm² (d) 168 cm²

- 64.** The radius of a circle is increased so that its circumference increases by 15%. The area of the circle will increase by

[CDS 2017(I)]

- (a) 31.25% (b) 32.25%
(c) 33.25% (d) 34.25%

- 65.** An isosceles triangle is drawn outside on one of the sides of a square as base in such a way that the perimeter of the complete figure is $\frac{7}{6}$ times the perimeter of the original square. What is the ratio of area of the triangle to the area of the original square?

[CDS 2017(II)]

- (a) 1:1 (b) 2:2
(c) 1:2 (d) 1:3

- 66.** What is the area of the triangle whose sides are 51 cm, 37cm and 20cm?

[CDS 2017(II)]

- (a) 300 square cm (b) 305 square cm
(c) 306 square cm (d) 307 square cm

- 67.** Two straight lines AB and AC include an angle. A circle is drawn in this angle which touches both these lines. One more circle is drawn which touches both these lines as well as the previous circle. If the area of the bigger circle is 9 times the area of the smaller circle, then what must be the angle A?

[CDS 2017(II)]

- (a) 45° (b) 60°
(c) 75° (d) 90°

- 68.** The diameters of two given circles are in the ratio 12:5 and the sum of their areas is equal to the area of a circle of diameter 65cm. What are their radii?

[CDS 2017(II)]

- (a) 12cm and 5cm
(b) 24cm and 10cm
(c) 60 cm and 25cm
(d) 30cm and 12.5cm

- 69.** Segment QR of length r is a tangent at Q to a circle of radius r with center at P. What is the area of the part of the triangle PQR, which is outside the circular region?

[CDS 2017(II)]

- (a) $\frac{\pi r^2}{16}$ (b) $\frac{r^2}{2} - \frac{\pi r^2}{8}$
(c) $\frac{r^2}{2} - \frac{\pi r^2}{16}$ (d) $\frac{r^2}{4} - \frac{\pi r^2}{8}$

- 70.** If the length of a side of a square is increased by 8cm, its area increases by 120 square cm. What is the length of a side of the square?

[CDS 2018(I)]

- (a) 2.5cm (b) 3.5cm
(c) 4.5cm (d) 5.5cm

- 71.** A rectangular pathway having width 4.5m and length 10m will have to be tiled using square tiles of side 50cm. Each packet of such tiles contains 20 pieces and costs Rs100. What will be the total cost of tiles for the pathway?

[CDS 2018(I)]

- (a) Rs 1,200 (b) Rs 1,100
(c) Rs 1,000 (d) Rs 900

- 72.** The product of the length of the diagonals of a square is 50 square units. What is the length of a side of the square?

[CDS 2018(I)]

- (a) $5\sqrt{2}$ units (b) 5 units
(c) 10 units (d) $2\sqrt{5}$ units

- 73.** Two equal circular regions of greatest possible area are cut off from a given circular sheet of area A. What is the remaining area of the sheet?

[CDS 2018(I)]

- (a) A/2 (b) A/3
(c) 3A/5 (d) 2A/5

- 74.** A square and an equilateral triangle have the same perimeter. If the diagonal of the square is $6\sqrt{2}$ cm, the what is the area of the triangle?

[CDS 2018(I)]

- (a) $12\sqrt{2}$ cm² (b) $12\sqrt{3}$ cm²
(c) $16\sqrt{2}$ cm² (d) $16\sqrt{3}$ cm²

- 75.** What is the area of the region bound externally by a square if side of length 'a' and internally by a square of side of length 'a' and internally by a circle passing through the four corners of the square?

[CDS 2018(I)]

- (a) $(\pi - 1)a^2$ square units
(b) $\frac{(\pi-1)a^2}{2}$ square units
(c) $(\pi - 2)a^2$ square units
(d) $\frac{(\pi-2)a^2}{2}$ square units

- 76.** A region of area A bounded by a circle C is divided into n regions, each of area A/n by drawing circle of radii $r_1, r_2, r_3, \dots, r_{n-1}$ such that $r_1 < r_2 < r_3 < \dots < r_{n-1}$ concentric with the circle C. If $p_m = \frac{r_{m+1}}{r_m}$ where $m = 1, 2, 3, \dots, (n-2)$, then which one of the following is correct?

[CDS-2018-1]

- (a) p increases as m increases
(b) p decreases as m increases
(c) p remains constant as m increases
(d) p increases for some values of m as m increases and then decreases thereafter

- 77.** A wire is in the form of a circle of radius 98cm. A square is formed out of the wire. What is the length of a side of the square? (Use $\pi = 22/7$)

[CDS-2018-1]

- (a) 146cm (b) 152cm
(c) 154cm (d) 156cm

- 78.** What is the area of the largest circular disc cut from a square of side $\frac{2}{\sqrt{\pi}}$ units?

[CDS-2018-1]

- (a) π square units (b) 1 square unit
(c) π^2 square units (d) 2 square units

- 79.** If base and hypotenuse of a right triangle are $(u^2 - v^2)$ and $(u^2 + v^2)$ respectively and the area of the triangle is 2016 square units, then the perimeter of the triangle may be:

[CDS 2018(II)]

- (a) 224 units (b) 288 units
(c) 448 units (d) 576 units

- 80.** Let S be the parallelogram obtained by joining the mid points of the parallelogram T. Consider the following statements:

1. The ratio of area of T to that of S is 2 : 1
2. The perimeter of S is half of the sum of diagonals T

Which of the above statements is/are correct?

[CDS 2018(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 81.** The sides of a triangle are 5 cm, 6 cm and 7 cm. The area of the triangle is approximately:

[CDS 2018(II)]

- (a) 14.9cm² (b) 14.7cm²
(c) 14.5 cm² (d) 14.3 cm²

- 82.** There is a path of width 5 m around a circular plot of land whose area is 144π m². The total area of the circular plot including the path surrounding it is:

[CDS 2018(II)]

- (a) 349m² (b) 289m²
(c) 209m² (d) 149π m²

- 83.** An equilateral triangle, a square and a circle have equal perimeter. If T, S and C denote the area of the triangle, area of the square and area of the circle respectively, then which one of the following is correct?

[CDS 2018(II)]

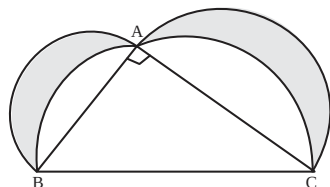
- (a) $T < S < C$ (b) $S < T < C$
(c) $C < S < T$ (d) $T < C < S$

- 84.** The area of the region bounded externally by a square of side 2a cm and internally by the circle touching the four sides of the square is:

[CDS 2018(II)]

- (a) $(4-\pi)a^2$ (b) $(\pi-2)a^2$
(c) $(8-\pi)a^2/2$ (d) $(\pi-2)a^2/2$

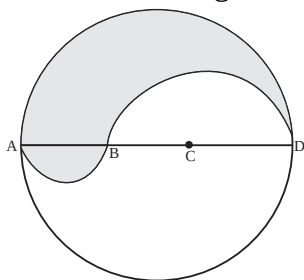
- 85.** In the figure given below, ABC is a right-angled triangle where $\angle A = 90^\circ$, $AB = p$ cm and $AC = q$ cm. On the three sides as diameters semicircles are drawn as shown in the figure. The area of the shaded portion, in square cm, is:



[CDS 2018(II)]

- (a) pq (b) $\pi(p^2 + q^2)/2$
(c) $\pi(p^2 + q^2)$ (d) $pq/2$

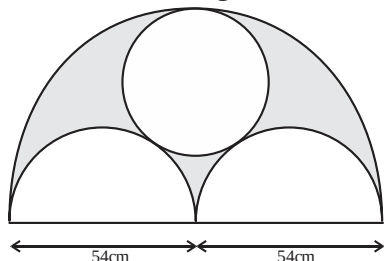
- 86.** In the figure given below, ABCD is the diameter of a circle of radius 9 cm. The lengths AB, BC and CD are equal. Semicircles are drawn on AB and BD as diameters as shown in the figure. What is the area of the shaded region?



[CDS 2018(II)]

- (a) 9π (b) 27π
(c) 36π (d) 81π

- 87.** In the figure given below, the diameter of bigger semicircle is 108cm. What is the area of the shaded region?

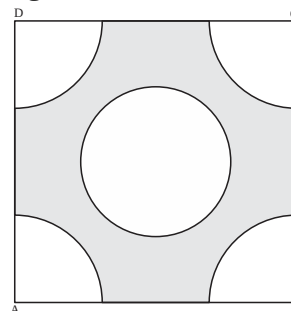


[CDS 2018(II)]

- (a) $201\pi\text{cm}^2$ (b) $186.3\pi\text{cm}^2$
(c) $405\pi\text{cm}^2$ (d) $769.5\pi\text{cm}^2$

- 88.** In the figure given below, ABCD is a square of side 4 cm. Quadrants of a circle of diameter 2 cm are removed from the four

corners and a circle of diameter 2 cm is also removed. What is the area of the shaded region?



[CDS 2018(II)]

- (a) $5\frac{7}{9}\text{cm}^2$ (b) $7\frac{7}{9}\text{cm}^2$
(c) $9\frac{5}{9}\text{cm}^2$ (d) $9\frac{5}{6}\text{cm}^2$

- 89.** In a rectangle, length is three times its breadth. If the length and the breadth of the rectangle are increased by 30% and 10% respectively, then its perimeter increases by :

[CDS 2019(I)]

- (a) $\frac{40}{3}\%$ (b) 20%
(c) 25% (d) 27%

- 90.** What is the percentage decrease in the area of a triangle if its each side is halved?

[CDS 2019(I)]

- (a) 75% (b) 50%
(c) 25% (d) No change

- 91.** Considering two opposite vertices of square of side 'a' as centres, two circular area are drawn within the square joining the other two vertices, thus forming two sectors. What is the common area in these two sectors?

[CDS 2019(I)]

- (a) $a^2 \left(\pi + \frac{1}{2} \right)$ (b) $a^2 \left(\pi - \frac{1}{2} \right)$
(c) $a^2 \left(\frac{\pi}{2} - 1 \right)$ (d) $a^2 \left(\frac{\pi}{2} + 1 \right)$

- 92.** A rectangular red carpet of size 6ft $\times$ 12 ft has a dark red border 6 inches wide. What is the area of the dark red border?

[CDS 2019(I)]

- (a) 9 square feet (b) 15 square feet
(c) 17 square feet (d) 18 square feet

93. A 12 m long wire is cut into two pieces, one of which is bent into a circle and the other into a square enclosing the circle. What is the radius of the circle?

[CDS 2019(I)]

- (a) $\frac{12}{\pi+4}$ (b) $\frac{6}{\pi+4}$
(c) $\frac{3}{\pi+4}$ (d) $\frac{6}{\pi+2\sqrt{2}}$

94. The hypotenuse of a right-angled triangle is 10 cm and its area is 24 cm². If the shorter side is halved and the longer side is doubled, the new hypotenuse becomes:

[CDS 2019(I)]

- (a) $\sqrt{245}$ cm (b) $\sqrt{255}$ cm
(c) $\sqrt{265}$ cm (d) $\sqrt{275}$ cm

Direction (for next three): ABCD is a quadrilateral with AB = 9cm, BC = 40cm, CD = 28cm, DA = 15 cm and angle ABC is a right angle.

95. What is the area of triangle ADC?

[CDS 2019(I)]

- (a) 126cm² (b) 124cm²
(c) 122cm² (d) 120cm²

96. What is the area of quadrilateral ABCD?

[CDS 2019(I)]

- (a) 300cm² (b) 306cm²
(c) 312cm² (d) 316cm²

97. What is the difference between perimeter of triangle ABC and perimeter of triangle ADC?

[CDS 2019(I)]

- (a) 4cm (b) 5cm
(c) 6cm (d) 7cm

98. The perimeter of a triangle is 22cm. Through each vertex of the triangle, a straight line parallel to the opposite side is drawn. What is the perimeter of triangle formed by these lines?

[CDS 2019(II)]

- (a) 33cm (b) 44cm
(c) 66cm (d) 88cm

99. The sides AD, BC of a trapezium ABCD are parallel and the diagonals AC and BD meet O. If the area of triangle AOB is 3 cm² and the area of triangle BDC is 8cm², then what is the area of triangle AOD?

[CDS 2019(II)]

- (a) 8cm² (b) 5cm²
(c) 3.6cm² (d) 1.8cm²

100. An equilateral triangle and a square are constructed using metallic wires of equal length. What is the ratio of area of triangle to that of square?

[CDS 2019(II)]

- (a) 3:4 (b) 2:3
(c) $4\sqrt{3}:9$ (d) $2\sqrt{3}:9$

101. All the four sides of a parallelogram are of equal length. The diagonals are in the ratio 1:2. If the sum of the lengths of the diagonals is 12cm, then what is the area of the parallelogram?

[CDS 2019(II)]

- (a) 9cm² (b) 12cm²
(c) 16cm² (d) 25cm²

102. The length and breadth of a rectangle are increased by 20% and 10% respectively. What is the percentage increase in the area of the rectangle?

[CDS 2019(II)]

- (a) 32% (b) 30%
(c) 25% (d) 15%

103. A square is drawn such that its vertices are lying on a circle of radius 201 mm. What is the ratio of area of circle to that of square?

[CDS 2019(II)]

- (a) 11:7 (b) 7:11
(c) 20:19 (d) 19:20

104. What is the ratio of the area of a square inscribed in a semicircle of radius r to the area of square inscribed in a circle of radius r?

[CDS 2019(II)]

- (a) 1:2 (b) 2:5
(c) 2:3 (d) 3:5

105. The area of a sector of a circle of radius 4cm is 25.6cm^2 . What is the radian measure of the arc of the sector?

[CDS 2019(II)]

- (a) 2.3 (b) 3.2
(c) 3.3 (d) 3.4

106. What is the area of the triangle having side lengths $\frac{y}{z} + \frac{z}{x}, \frac{z}{x} + \frac{x}{y}, \frac{x}{y} + \frac{y}{z}$?

[CDS 2020(I)]

- (a) $\frac{(x+y+z)^2}{xyz}$ (b) $\frac{\sqrt{xyz}}{x+y+z}$
(c) $\sqrt{\frac{x}{y} + \frac{y}{z} + \frac{z}{x}}$ (d) $\sqrt{\frac{xy+yz+zx}{xyz}}$

107. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3}+1)\text{cm}$, then what is the area of the triangle?

[CDS 2020(I)]

- (a) $(\sqrt{3}+1)\text{cm}^2$ (b) $(\sqrt{3}+3)\text{cm}^2$
(c) $\frac{1}{2}(\sqrt{3}+1)\text{cm}^2$ (d) $2(\sqrt{3}+1)\text{cm}^2$

108. Two circles touch internally. The sum of their areas is $136\pi\text{ cm}^2$ and distance between their centres is 4cm. What are the radii of the circles?

[CDS 2020(I)]

- (a) 11cm, 7cm (b) 10cm, 6cm
(c) 9cm, 5cm (d) 8cm, 4cm

109. If area of a circle and a square are same, then what is the ratio of their perimeters?

[CDS 2020(I)]

- (a) $2\sqrt{\pi}$ (b) $\sqrt{\pi}$
(c) $\frac{\sqrt{\pi}}{2}$ (d) $\frac{\sqrt{\pi}}{4}$

110. A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the

length of the side of the square then the other side of the rectangle is:

[CDS 2020(I)]

- (a) Square root of the side of the square
(b) Half the side of the square
(c) Of unit length
(d) Double the side of the square

111. The length and breadth of a rectangle are in the ratio 4:3. Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side?

[CDS 2020(I)]

- (a) 3:4 (b) 4:3
(c) 16:9 (d) 1:1

112. Suppose a region is formed by removing a sector of 20° from a circular region of radius 30 feet. What is the area of this new region?

[CDS 2020(I)]

- (a) 150π square feet (b) 550π square feet
(c) 650π square feet (d) 850π square feet

113. If the diagonals of a rhombus are x and y, then what is its area?

[CDS 2020(I)]

- (a) $\frac{xy}{2}$ (b) $\frac{xy}{4}$
(c) xy (d) x^2-y^2

114. What is the area of the largest square plate cut from a circular disk of radius one unit?

[CDS 2020(I)]

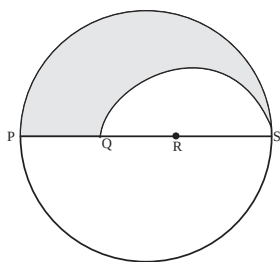
- (a) 4 square units (b) $2\sqrt{2}$ square units
(c) π square units (d) 2 square units

115. ABCD is a quadrilateral such that $AD = DC = CA = 20$ units, $BC = 12$ units and $\angle ABC = 90^\circ$. What is the approximate area of the quadrilateral ABCD?

[CDS 2020(I)]

- (a) 269 sq. units (b) 300 sq. units
(c) 325 sq. units (d) 349 sq. units

116.

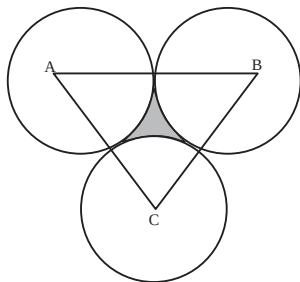


Let PQRS be the diameter of a circle of radius 9 cm. The length PQ, QR and RS are equal. Semi-Circle is drawn with QS as diameter (as shown in the given figure). What is the ratio of the shaded region to that of the unshaded region?

[CDS 2020(I)]

- (a) 25:121 (b) 5:13
(c) 5 : 18 (d) 1:2

117.

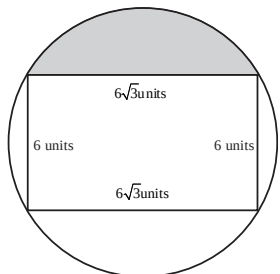


What is the area of the shaded region in the given figure, if the radius of each of the circles is 2cm?

[CDS 2020(I)]

- (a) $4\sqrt{3} - 2\text{cm}^2$ (b) $\sqrt{3} - \pi\text{cm}^2$
(c) $\sqrt{3} - \frac{\pi}{2}\text{cm}^2$ (d) $2\pi - 2\sqrt{3}\text{cm}^2$

118.



In the given figure, what is the area of the shaded region?

[CDS 2020(I)]

- (a) $9(\pi - \sqrt{3})\text{square units}$
(b) $3(4\pi - 3\sqrt{3})\text{square units}$

(c) $3(3\pi - 4\sqrt{3})\text{square units}$

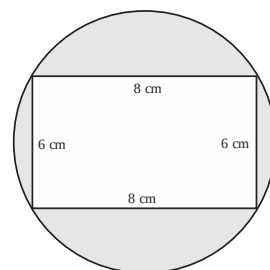
(d) $9(\sqrt{3} - \pi)\text{square units}$

119. ABCD is a trapezium, where AB is parallel to DC. If AB = 4cm, BC = 3cm, CD = 7cm and DA = 2cm, then what is the area of the trapezium?

[CDS 2020(I)]

- (a) $22\sqrt{\frac{2}{3}}\text{cm}^2$ (b) $22\sqrt{\frac{3}{2}}\text{cm}^2$
(c) $22\sqrt{3}\text{cm}^2$ (d) $\frac{22\sqrt{2}}{3}\text{cm}^2$

120.



What is the approximate area of the shaded region in the figure given?

[CDS 2020(I)]

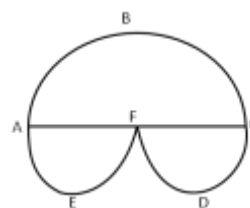
- (a) 15.3cm² (b) 25.5cm²
(c) 28.4cm² (d) 30.5cm²

121. A circle of diameter 8cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle?

[CDS 2020(I)]

- (a) $4(3 - \sqrt{2})\text{cm}$ (b) $4(3 - 2\sqrt{2})\text{cm}$
(c) $8(3 - \sqrt{2})\text{cm}$ (d) $8(3 - 2\sqrt{2})\text{cm}$

122.



In the given figure, there are three semi circle ABC, AEF and CDF. The distance

between A and C is 28 units and F is the midpoint of AC. What is the total area of the three semi circles?

[CDS 2020(I)]

- (a) 924 square units (b) 824 square units
(c) 624 square units (d) 462 square units

123. The two sides of a triangle are 40cm and 41cm. If the perimeter of the triangle is 90cm, what is its area?

[CDS 2020(II)]

- (a) 90 cm² (b) 135 cm²
(c) 150 cm² (d) 180 cm²

124. The diagonals of a rhombus differ by 2 units and its perimeter exceeds the sum of the diagonals by 6 units. What is the area of the rhombus?

[CDS 2020(II)]

- (a) 48 square units (b) 36 square units
(c) 24 square units (d) 12 square units

125. The length of a rectangle is increased by 10% and breadth is decreased by 10%. Then the area of new rectangle is

[CDS 2020(II)]

- (a) neither increased nor decreased
(b) increased by 1%
(c) decreased by 1%
(d) decreased by 10%

126. If the perimeter of a circle and square are equal, then what is the ratio of the area of the circle to that of the square?

[CDS 2020(II)]

- (a) 1 : π (b) 2 : π
(c) 3 : π (d) 4 : π

127. Areas of two squares are in the ratio $m^2:n^4$, what is the ratio of their perimeter?

[CDS 2020(II)]

- (a) $m : n$ (b) $n : m$
(c) $m : n^2$ (d) $m^2 : n$

128. Four circular coins of equal radius are placed with their centres coinciding four vertices of a square. Each coin touches two other coins. If the uncovered area of the square is 42 cm², then what is the radius of each coin

[CDS 2020(II)]

- (a) 5 cm (b) 7 cm
(c) 10 cm (d) 14 cm

129. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55cm, then what is the area of the sector?

[CDS 2021(I)]

- (a) 577.5cm² (b) 612.5 cm²
(c) 705.5 cm² (d) 725.5 cm²

130. A wire is in the form of a circle of radius 70 cm. If it is bent in the form of a rhombus, then what is its side length? $\left(Take \pi = \frac{22}{7} \right)$.

[CDS 2021(II)]

- (a) 55cm (b) 75cm
(c) 95cm (d) 110cm

131. If the perimeter of a semicircular park is 360m, then what is its area? $\left(Take \pi = \frac{22}{7} \right)$

[CDS 2021(II)]

- (a) 3850 m² (b) 7700m²
(c) 11550 m² (d) 15400 m²

132. The radius of circum-circle of a right angled triangle is 10cm and the altitude drawn to the hypotenuse is 8cm. What is the area of the triangle?

[CDS 2021(II)]

- (a) 60cm² (b) 80cm²
(c) 100cm² (d) 120cm²

133. Two circles touch externally. The sum of their areas is 41π square cm. If the distance between their centres is 9cm, then what is difference between their diameters?

[CDS 2021(II)]

- (a) 1cm (b) 1.5cm
(c) 2cm (d) 4cm

134. Let p be the area of a square X and q be the area of the square formed on the diagonal of the square X. What is the value of $\frac{p}{q}$?

[CDS 2021(II)]

- (a) $\frac{1}{8}$ (b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{2}$

- 135.** The area of a rhombus is 336 square cm. If the length of one of its diagonals is 48cm, then what is the perimeter of the rhombus?

[CDS 2021(II)]

- (a) 200cm (b) 120cm
(c) 100cm (d) 90cm

- 136.** The minute hand of a clock is 21cm long. What is the area on the face of the clock described by the minute hand between 10.10 a.m and 10.30 a.m.? (take $\pi = \frac{22}{7}$).

[CDS 2021(II)]

- (a) 231 cm² (b) 331 cm²
(c) 462 cm² (d) 492 cm²

Consider the following for the next two (02) items that follow:

A chord of a circle of radius 2.1 cm subtends an angle of 120° at the centre. (take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$)

- 137.** What is the approximate area of minor segment of the circle?

[CDS 2021(II)]

- (a) 2.71 cm² (b) 2.42 cm²
(c) 1.91 cm² (d) 1.71 cm²

- 138.** What is the approximate area of major segment of the circle?

[CDS 2021(II)]

- (a) 10.05 cm² (b) 10.15 cm²
(c) 11.05 cm² (d) 11.15 cm²

- 139.** A person wishes to fence 375 m² rectangular garden. He has 65m of barbed wire and is able to fence only three sides of the garden. What is the perimeter of the garden?

[CDS 2021(II)]

- (a) 80m (b) 84m
(c) 90m (d) 100m

- 140.** A bicycle wheel of radius 35cm makes n revolutions in moving 11km. What is the value of n? (take $\pi = \frac{22}{7}$)

[CDS 2022(I)]

- (a) 500 (b) 1000
(c) 2500 (d) 5000

- 141.** A farmland is in the shape of a rhombus. The perimeter of the land is 100 m and the

length of one of the diagonals is 40m. The land is divided into four equal parts. What is the area of each part?

[CDS 2022(I)]

- (a) 150 square meter
(b) 225 square meter
(c) 300 square meter
(d) 450 square meter

- 142.** If the perimeter of an isosceles right triangle is $4(2 + \sqrt{2})$ cm, then what is its area in square cm?

[CDS 2022(I)]

- (a) 8 (b) 12
(c) 16 (d) 24

- 143.** A rectangle of length 10 units and breadth 8 unit is split into two squares each of area x square units and two rectangles each of area y square units. Consider the following statements:

1.y is always greater than x

2.y can be 15 square units.

Which of the above statement is/are correct?

[CDS 2022(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 144.** A square sheet is formed by joining n identical square sheets of same size. If the length of the diagonal of the bigger square sheet so formed is m, then what is the side length of a smaller square sheet?

[CDS 2022(I)]

- (a) $\frac{m}{\sqrt{n}}$ (b) $\frac{m}{2\sqrt{n}}$
(c) $\frac{m}{\sqrt{2n}}$ (d) $\frac{\sqrt{2}m}{\sqrt{n}}$

- 145.** What is the area of the region enclosed by three identical circles (each of radius 4cm) touching each other?

[CDS 2022(I)]

- (a) $\frac{8\pi}{3}$ square cm
(b) $\left(16\sqrt{3} - \frac{8\pi}{3}\right)$ square cm
(c) $(16\sqrt{3} - 8\pi)$ square cm

(d) $\frac{16\pi}{\sqrt{3}}$ square cm

- 146.** X, Y and Z are three equilateral triangles. The sum of the areas of X and Y is equal to the area of Z. If the side lengths of X and Y are 6cm and 8cm respectively, then what is the side length of Z?

[CDS 2022 (II)]

- (a) 9cm (b) 9.5cm
(c) 10cm (d) 10.5cm

- 147.** What is the time taken by a person to cover one round of a circular park of diameter 210 m if he walks at a speed of 6km/hr? ($\pi = \frac{22}{7}$)

[CDS 2022 (II)]

- (a) 6.6 minutes (b) 5.5 minutes
(c) 4.4 minutes (d) 3.3 minutes

- 148.** In an equilateral triangle of side $2\sqrt{3}$ cm, a circle is inscribed touching the sides. What is the area of the remaining portion of the triangle?

[CDS 2022 (II)]

- (a) $(2\sqrt{3} - \pi)$ square cm
(b) $(3\sqrt{3} - \pi)$ square cm
(c) $(4\sqrt{3} - 2\pi)$ square cm
(d) $(4\sqrt{3} - \pi)$ square cm

- 149.** A triangle has side lengths x cm, x + 13cm and x + 26cm. If its area is 126 square cm, then what is the value of x?

[CDS 2022 (II)]

- (a) 18 (b) 17
(c) 16 (d) 15

- 150.** Two parallel sides of a trapezium are 29cm and 21cm. Non-parallel sides are equal and each is of length 8.5 cm. What is the area of the trapezium?

[CDS 2022 (II)]

- (a) 187.5 square cm
(b) 227.5 square cm
(c) 375 square cm
(d) 455 square cm

- 151.** A square copper plate of side 16cm weighs 128 gm. A circular disc of diameter 14 cm is cut off from the plate. What is the weight of the remaining part? ($\pi = \frac{22}{7}$)

[CDS 2022 (II)]

- (a) 48gm (b) 49gm
(c) 50gm (d) 51gm

- 152.** The area of a rhombus is 96 square cm and one of its diagonals is of length 12cm. What is the perimeter of the rhombus?

[CDS 2022 (II)]

- (a) 36cm (b) 40cm
(c) 44cm (d) 48cm

- 153.** Two circles touch externally. The sum of their areas is 89π square cm and the distance between their centres is 13cm. What is the difference in their radii?

[CDS 2022 (II)]

- (a) 2cm (b) 2.5cm
(c) 3cm (d) 3.5cm

- 154.** A square and a rectangle have same perimeter. They differ in areas by 1 square cm. The length of the rectangle exceeds its breadth by:

[CDS 2022 (II)]

- (a) 1cm (b) 2cm
(c) 3cm (d) 4cm

- 155.** Two rectangles are of same area equal to 480 square cm. They differ in lengths by 6cm and breadths by 4cm. What is the difference in their perimeters?

[CDS 2022 (II)]

- (a) 2cm (b) 4cm
(c) 6cm (d) 10cm

Consider the following for the next two (02) items that follow:

A chord of length l of a circle makes an angle 90° at the centre of the circle.

- 156.** What is the area of the minor segment?

[CDS 2022 (II)]

- (a) $\frac{l^2}{2} \left(\pi - \frac{1}{2} \right)$ (b) $\frac{l^2}{4} \left(\pi - \frac{1}{2} \right)$
(c) $\frac{l^2}{4} \left(\frac{\pi}{2} - 1 \right)$ (d) $\frac{l^2}{2} \left(\frac{\pi}{2} - \frac{1}{2} \right)$

- 157.** What is the area of the major segment?

[CDS 2022 (II)]

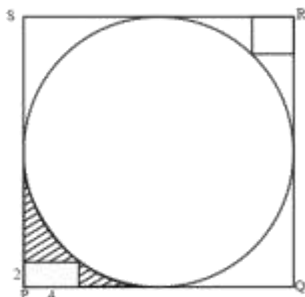
- (a) $\frac{l^2}{4} \left(\frac{3\pi}{2} + 1 \right)$ (b) $\frac{l^2}{4} \left(\frac{3\pi}{2} - 1 \right)$

(c) $\frac{l^2}{2} \left(\frac{3\pi}{2} + 1 \right)$

(d) $\frac{l^2}{2} \left(\frac{3\pi}{2} - 1 \right)$

Consider the following for the next three (03) items that follow:

In the figure given below, a circle is inscribed in a square PQRS. A rectangle at the corner P that measures 4cm $\times$ 2cm and a square at the corner R are drawn.



158. What is the area of the circle?

[CDS – 2023 (1)]

- (a) 100π square cm (b) 96π square cm
(c) 50π square cm (d) 48π square cm

159. What is the area of the smaller square?

[CDS – 2023 (1)]

- (a) $50(3 - \sqrt{2})$ square cm
(b) $25(3 - 2\sqrt{2})$ square cm
(c) $25(3 + 2\sqrt{2})$ square cm
(d) $50(3 - 2\sqrt{2})$ square cm

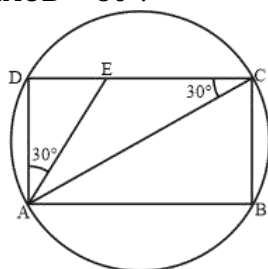
160. What is the area of the shaded region?

[CDS – 2023 (1)]

- (a) $(96 - 25\pi)$ square cm
(b) $(92 - 25\pi)$ square cm
(c) $(96 - 16\pi)$ square cm
(d) $(92 - 16\pi)$ square cm

Consider the following for the next two (02) items that follow:

In the following figure, a rectangle ABCD is inscribed in a circle of radius r. given $\angle DAE = 30^\circ$ and $\angle ACD = 30^\circ$.



161. What is the ratio of the area of the circle to the area of the rectangle?

[CDS – 2023 (1)]

(a) $\frac{\pi}{\sqrt{2}}$

(b) $\frac{\pi}{\sqrt{3}}$

(c) $\frac{2\pi}{\sqrt{3}}$

(d) $\frac{3\pi}{\sqrt{2}}$

162. What is the area of $\triangle AEC$?

[CDS – 2023 (1)]

(a) $\frac{r^2}{\sqrt{3}}$

(b) $\frac{r^2}{2\sqrt{3}}$

(c) $\frac{r^2}{3\sqrt{3}}$

(d) $\frac{3r^2}{\sqrt{3}}$

163. The plinth of a house has an area of 200 square metres. It is rectangular in shape and its length and breadth are in the ratio 2:1. The owner of the house extends the terrace by 1 m on each side. What is the percentage of area that has increased in the terrace relative to the plinth?

[CDS – 2023 (1)]

- (a) 40% (b) 32%
(c) 20% (d) 15.5%

164. A square sheet of side length 44 cm is rolled along one of its sides to form a cylinder by making opposite edges just to touch each other. What is the volume of the cylinder? (Take $\pi = \frac{22}{7}$).

[CDS – 2023 (1)]

- (a) 6776 cubic cm (b) 6248 cubic cm
(c) 5896 cubic cm (d) 5680 cubic cm

165. The perimeter and the area of a right-angled triangle are 36 cm and 54 square cm respectively. What is the length of the hypotenuse?

[CDS – 2023 (1)]

- (a) 12cm (b) 14cm
(c) 15cm (d) 16cm

166. The perimeter of a sector of a circle of radius 5.2cm is 16.4 cm. What is the area of the sector?

[CDS – 2023 (1)]

- (a) 15.6 square cm (b) 15 square
(c) 14.4 square cm (d) 14.1 square cm

167. What is the area of the circle (approximately) inscribed in a triangle with side lengths 12cm, 16cm and 20cm?

[CDS – 2023 (1)]

- (a) 48 square cm (b) 50 square cm
(c) 52 square cm (d) 54 square cm

168. A floor of a big hall has dimensions 30m 60cm and 23m 40cm. Its is to be paved with square tiles of same size. What is the minimum number of tiles required?

[CDS – 2023 (1)]

- (a) 30 (b) 36
(c) 169 (d) 221

169. How long will a man take to walk around the boundary of a square field of area 25 hectares at the rate of 5km/hr?

[CDS – 2023 (1)]

- (a) 36 minutes (b) 30 minutes
(c) 24 minutes (d) 18 minutes

170. Let x be the area of a square inscribed in a circle of radius r and y be the area of an equilateral triangle inscribed in the same circle. Which one of the following is correct?

[CDS – 2023 (1)]

- (a) $9x^2 = 16y^2$ (b) $27x^2 = 64y^2$
(c) $36x^2 = 49y^2$ (d) $16x^2 = 21y^2$

171. If the length of a rectangle is increased by $66\frac{2}{3}\%$, then by what percent should the width of the rectangle be decreased in order to maintain the same area?

[CDS – 2023 (1)]

- (a) 50% (b) 45%
(c) 40% (d) 35%

172. What is the maximum area that can be covered by three non-intersecting circles drawn inside a rectangle of sides 8 cm and 12cm?

[CDS – 2023 (1)]

- (a) 16π square cm (b) 18π square cm
(c) 20π square cm (d) 24π square cm

173. A triangle ABC has been divided into four smaller triangles P, Q, R, S whose perimeters are 16cm, 12cm, 4cm and 12cm respectively. P, R and S contain the vertices A, B and C respectively. What is the perimeter of the triangle ABC?

[CDS – 2023 (1)]

- (a) 18cm (b) 20cm
(c) 22cm (d) 24cm

174. OABC is a rhombus whose three vertices lie on a circle with centre at O. If the area of the rhombus is $32\sqrt{3}$ square cm, then what is the radius of the circle?

[CDS – 2023 (1)]

- (a) 4cm (b) 6cm
(c) 8cm (d) 16cm

175. Question: What is the area of the circle C?

Statement I: an arc of length 7 cm subtends an angle 30° at the centre of C.

Statement II: A chord of length 10cm subtends an angle 90° at the centre of C.

[CDS – 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together.

176. A chord PQ of the circle C divides it into two segments such that 3 times the area of the major segment is 4 times the area of the minor segment.

Question: What is the radius of C?

Statement I: Area of the minor segment is 66 square cm.

Statement II: Area of the major segment is 88 square cm.

[CDS – 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together.

177. An isosceles triangle has its base length $2a$ and its height is h . On each side of the triangle a square is drawn external to the triangle. What is the area of the figure thus formed?

[CDS-2023 (2)]

- (a) $6a^2 + 2h^2 + 2ah$ (b) $6a^2 + 2h^2 + ah$
(c) $4a^2 + 2h^2 + ah$ (d) $6a^2 + h^2 + ah$

178. A pendulum swings through an angle of 9° and its end describes an arc of length 14.3cm. What is the length of the pendulum? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 88cm (b) 91cm
(c) 95cm (d) 98cm

179. The corners of an equilateral triangular plate were cut in such a manner that it forms a regular hexagonal plate. What is the ratio of the area of the triangular plate to the area of the hexagonal plate?

[CDS-2023 (2)]

- (a) 2:1 (b) 3:2
(c) 4:3 (d) 5:3

180. Two equal arcs of different circles C_1 and C_2 subtend angles of 60° and 75° respectively, at the centres. What is the ratio of the radius of C_1 to the radius of C_2 ?

[CDS-2023 (2)]

- (a) 4:5 (b) 5:4
(c) 1:1 (d) 3:2

181. ABC is a triangle with sides $AB = 41$ cm, $BC = 28$ cm and $CA = 15$ cm. If D, E and F are the mid points of AB, BC and CA respectively, then what is the area of the triangle DEF?

[CDS-2023 (2)]

- (a) 63 square cm (b) 45 square cm
(c) 31.5 square cm (d) 22.5 square cm

182. The circumference of a circle exceeds the diameter by 16.8 cm. What is the diameter of the circle? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 6.24cm (b) 6.42cm
(c) 7.64cm (d) 7.84cm

183. What is the area of the region between the concentric circles if the chord of the out circle of length 14cm is a tangent of the inn circle? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 125 square cm (b) 132 square cm
(c) 144 square cm (d) 154 square cm

Consider the following for the next two (02) items that follow:

A line segment AB is bisected at C and semi circles S_1 , S_2 and S_3 are drawn respectively on AB, AC and CB as diameters such that they all lie on same side of AB. A circle S is drawn touching internally S_1 and externally S_2 and S_3 .

184. If r is the radius of S and R is the radius of S_2 , then which one of the following is correct?

[CDS-2023 (2)]

- (a) $R=3r$ (b) $R=2r$
(c) $3R=4R$ (d) $2R=3r$

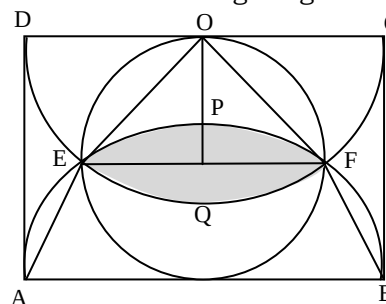
185. If m is the area of the circle S and n is the area of semi-circle S_1 , then which one of the following is correct?

[CDS-2023 (2)]

- (a) $9m=2n$ (b) $9m=4n$
(c) $3m=2n$ (d) $7m=3n$

Consider the following for the next three (03) items that follow:

Consider two identical semicircles and one circle inscribed in a rectangle of length 10 cm as shown in the figure given below.



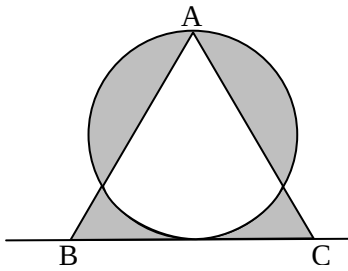
(Take $\pi = 3.14$ and $\sqrt{2} = 1.4$)

[CDS-2024 (1)]

186. What is the area of triangle EOF?
(a) $12.5\sqrt{3}$ square cm (b) $6.25\sqrt{3}$ square cm
(c) 12.5 square cm (d) 6.25 square cm
187. What is the area of trapezium AEFB?
(a) 30 square cm (b) 25 square cm
(c) 20 square cm (d) 18.75 square cm
188. What is the area of the shaded region?
(a) 14.75 square cm (b) 14.25 square cm
(c) 7.225 square cm (d) 7.625 square cm

Consider the following for the next two (02) items that follow:

Consider a circle of area 9π square unit and an equilateral triangle ABC as shown in the figure given below.



[CDS-2024 (1)]

189. What is the length of the side of the triangle ABC?

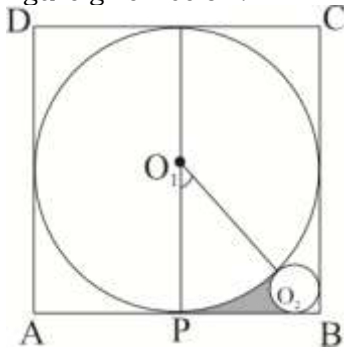
- (a) $2\sqrt{3}$ unit (b) $4\sqrt{3}$ unit
(c) $6\sqrt{3}$ unit (d) $8\sqrt{3}$ unit

190. What is the area of the shaded region?

- (a) $6(\pi + \sqrt{3})$ square unit
(b) $3(\pi + 2\sqrt{3})$ square unit
(c) $1.5(3\pi + 8\sqrt{3})$ square unit
(d) $6(\pi + 2\sqrt{3})$ square unit

Consider the following for the next three (03) items that follow:

Two circles with centres at O_1 and O_2 touching each other are placed inside a rectangle of sides 9 cm and 8 cm as shown in the figure given below:



[CDS-2024 (1)]

191. What is the sum of the areas of the two circles?

- (a) 17π square unit
(b) 16.75π square unit
(c) 16.5π square unit
(d) 16.25π square unit

192. Which one of the following is correct in respect of angle θ ?

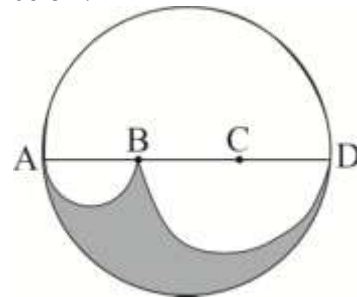
- (a) $0^\circ < \theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

193. What is the area of the shaded region?

- (a) $\frac{240 - 10\pi - \pi\theta}{24}$ square unit
(b) $\frac{240 - 6\pi - \pi\theta}{24}$ square unit
(c) $\frac{120 - 12\pi - \pi\theta}{24}$ square unit
(d) $\frac{240 - 12\pi - \pi\theta}{24}$ square unit

Consider the following for the next two (02) items that follow:

Let ABCD be the diameter of a circle of radius 6 cm. The lengths AB, BC and CD are equal. Semi-circles are drawn with AB and BD as diameters as shown in the figure given below.



[CDS-2024 (1)]

194. What is the ratio of the area of the shaded region to that of the non-shaded region?

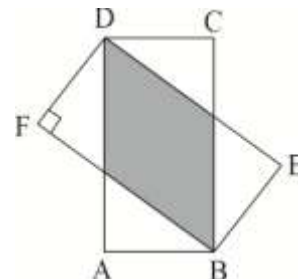
- (a) 2:7 (b) 2:5
(c) 3:5 (d) 5:8

195. What is the perimeter of the shaded region?

- (a) 24π cm (b) 18π cm
(c) 15π cm (d) 12π cm

Consider the following for the next two (02) items that follow:

Consider the two identical rectangles ABCD and BEDF as shown in the figure given below. Let AB = 1 cm and BC = 2 cm.



[CDS-2024 (1)]

196. What is the area of the overlapping region?

- (a) $\frac{8}{5}$ square cm (b) $\frac{5}{4}$ square cm

- (c) $\frac{4}{5}$ square cm (d) $\frac{3}{4}$ square cm

197. What is the area of the non-overlapping region?

- (a) $\frac{3}{4}$ square cm (b) $\frac{11}{4}$ square cm
(c) $\frac{3}{2}$ square cm (d) $\frac{5}{4}$ square cm

Consider the following for the next two (02) items that follow:

In a pie-diagram (with radius 7 cm), the central angles of the sectors are in the ratio 2 : 3 : 7 : 5 : 1.

(Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

198. If P is the area of the smallest sector and Q is the area of the largest sector, then what is P + Q equal to?

- (a) $88/3$ square cm (b) $77/3$ square cm
(c) $149/6$ square cm (d) $616/9$ square cm

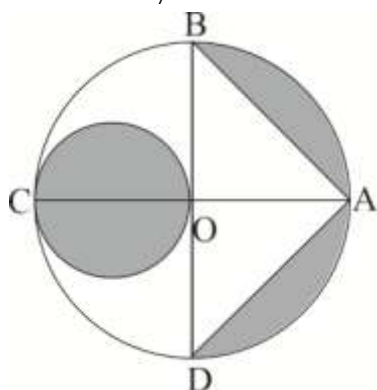
199. If p is the perimeter of the smallest sector, then what is the value of 9p?

- (a) 142 cm (b) 148 cm
(c) 156 cm (d) 221 cm

Consider the following for the next two (02) items that follow:

ABCD is a circle with centre O and taking OC as a diameter, a circle is drawn as shown in the figure given below. Let OB = 7

cm. (use $\pi = \frac{22}{7}$)



200. What is the area of the shaded region?

- (a) 38.5 square cm (b) 48 square cm
(c) 52.5 square cm (d) 66.5 square cm

201. What is the ratio of the area of the shaded region to the area of the non-shaded region?

- (a) $\frac{19}{25}$ (b) $\frac{18}{25}$
(c) $\frac{17}{25}$ (d) $\frac{16}{25}$

202. The chord AB of a circle with centre at O is $2\sqrt{3}$ times the height of the minor segment. If P is the area of the sector OAB and Q is the area of the minor segment of the circle, then what is the approximate value of $\frac{P}{Q}$?

(Take $\sqrt{3} = 1.7$ and $\pi = 3.14$)

[CDS-2024 (1)]

- (a) 1.4 (b) 1.7
(c) 2.2 (d) 2.6

203. In a right-angled triangle ABC, AB = 15 cm, BC = 20 cm and AC = 25 cm. Further, BP is the perpendicular on AC. What is the difference in the area of triangles PAB and PCB?

[CDS-2024 (1)]

- (a) 40 square cm (b) 42 square cm
(c) 45 square cm (d) 48 square cm

204. What is the area of the region between two concentric circles, if the length of a chord of the outer circle touching the inner circle at a particular point of its circumference is 14 cm? (Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

- (a) 154 square cm
(b) 144 square cm
(c) 132 square cm
(d) Cannot be determined due to insufficient data

205. ABC is a right-angled triangle, right-angled at B such that AB = 6 cm and BC = 8 cm. What is the perimeter of the square inscribed in the triangle ABC with maximum area?

[CDS-2024 (1)]

- (a) $24/7$ cm (b) $96/7$ cm
(c) 24 cm (d) 32 cm

206. Area of a rectangle with length x and breadth y is P and area of a parallelogram (which is strictly not a rectangle) with adjacent sides of length x and y is Q. Question: Is $P > Q$?

Statement-I: $x : y = 2 : 1$

Statement-II: The angle between the two adjacent sides of the parallelogram is 60°

[CDS-2024 (1)]

(a) If the Question can be answered by one of the Statements alone, but not by the other.

(b) If the Question can be answered by either Statement alone

(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.

(d) If the Question can be answered even by using both Statements together

ANSWER KEY

1.	c	2.	d	3.	c	4.	c	5.	c	6.	d	7.	c	8.	a	9.	b	10.	b
11.	a	12.	a	13.	a	14.	a	15.	a	16.	c	17.	b	18.	a	19.	b	20.	c
21.	d	22.	d	23.	b	24.	c	25.	c	26.	c	27.	c	28.	b	29.	c	30.	c
31.	a	32.	c	33.	d	34.	b	35.	b	36.	c	37.	b	38.	b	39.	a	40.	d
41.	d	42.	d	43.	d	44.	c	45.	b	46.	c	47.	d	48.	c	49.	a	50.	b
51.	a	52.	a	53.	b	54.	d	55.	b	56.	c	57.	b	58.	d	59.	b	60.	b
61.	d	62.	c	63.	c	64.	b	65.	d	66.	c	67.	b	68.	d	69.	b	70.	b
71.	b	72.	a	73.	a	74.	d	75.	d	76.	b	77.	c	78.	b	79.	b	80.	a
81.	b	82.	b	83.	a	84.	a	85.	d	86.	c	87.	c	88.	c	89.	c	90.	a
91.	c	92.	c	93.	b	94.	c	95.	a	96.	b	97.	c	98.	b	99.	d	100.	c
101.	c	102.	a	103.	a	104.	b	105.	b	106.	c	107.	c	108.	b	109.	c	110.	c
111.	d	112.	d	113.	a	114.	d	115.	a	116.	b	117.	a	118.	b	119.	d	120.	d
121.	b	122.	d	123.	d	124.	c	125.	c	126.	d	127.	c	128.	b	129.	a	130.	d
131.	b	132.	b	133.	c	134.	d	135.	c	136.	c	137.	a	138.	d	139.	a	140.	d
141.	a	142.	a	143.	b	144.	c	145.	c	146.	c	147.	a	148.	b	149.	d	150.	a
151.	d	152.	b	153.	c	154.	b	155.	b	156.	c	157.	a	158.	a	159.	d	160.	b
161.	b	162.	a	163.	b	164.	a	165.	c	166.	a	167.	b	168.	d	169.	c	170.	b
171.	c	172.	d	173.	b	174.	c	175.	b	176.	b	177.	b	178.	b	179.	b	180.	b
181.	c	182.	d	183.	d	184.	d	185.	a	186.	c	187.	a	188.	b	189.	b	190.	*
191.	a	192.	c	193.	d	194.	a	195.	d	196.	b	197.	c	198.	d	199.	b	200.	d
201.	a	202.	b	203.	b	204.	a	205.	b	206.	b								